Medical Microbiology

Lecture Three

Antibiotics

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Antibiotics:

An antibiotic is a chemical compound produced by certain microorganisms that can inhibit the growth of other microbes, even at low concentrations.

Discovery of Antibiotics:

- Discovered by: Alexander Fleming in 1928
- While returning from a holiday, he observed mold growing on his Staphylococcus agar plates.
- He noticed that bacteria did not grow near the mold, indicating the mold was inhibiting bacterial growth.
- This mold was later identified as Penicillium, leading to the discovery of penicillin, the first true antibiotic.

❖Penicillin the first Antibiotic:

- The bacterium on Fleming’s petri dish was Staphylococcus aureus.
- The mold that inhibited its growth belonged to the genus Penicillium.
- Penicillin was further developed by **Howard Florey** and **Ernst Chain** and became widely used during **World War II.**

❖Why do we use Antibiotics?

- To treat infections caused by bacteria.
- To prevent infections (prophylactic use, e.g., before surgeries) and more.

Antibiotic include:

- **Chemotherapy:** The use of chemical agents to treat diseases.
- **Antimicrobial Drugs**: Substances that inhibit the growth of microorganisms within a host.
- **Selective Toxicity:** The ability of a drug to target harmful microorganisms without harming the host.

Sources of Antibiotics:

Antibiotics primarily come from three major sources:

1-Fungi (molds).

2-Bacteria, especially: Streptomyces, Bacillus.

3-Synthetic or semi-synthetic compounds.

These sources characteristic by the following:

- Used internally or topically.
- Most effective against actively growing microorganisms.
- Designed to inhibit or kill pathogens.

Bacteriostatic verses Bactericidal:

Static: mean **inhibit** growth of M. O. (e.g. Tetracycline, Chloramphenicol).

Cidal: mean **kill** M.O. (e.g., ampicillin, gentamycin).

Cidal or static is not absolute, depending on concentration of drug, type of bacteria, phase of growth microorganism, and also the number of bacteria.

Bacteriostatic Activity:

- Bacteriostatic activity refers to the ability of an antimicrobial agent to **inhibit the growth** of microorganisms **without** necessarily **killing** them.
- This activity is measured **in vitro** by exposing a standardized number of organisms to varying concentrations of the antimicrobial agent.

- The **lowest concentration** that successfully prevents visible growth of the microorganism is called the **Minimum Inhibitory Concentration (MIC)**.
- When the antibiotic is stopped, the bacteria can start growing again.
- This type of antibiotic need the body’s immune system, like white blood cells, to remove and kill the bacteria.

Bactericidal activity :

-Is the ability of an antibiotic to **kill** bacteria directly, leading to a reduction in the number of viable bacterial cells.

-This is different from bacteriostatic activity, which only inhibits bacterial growth.

-Bactericidal antibiotics are often preferred in severe or life-threatening infections, especially in patients with weak immune systems.

Antibacterial spectrums:

Broad verses narrow spectrum:

- Antibiotics are active against several types of microorganisms are said to be broad spectrum

eg., tetracycline are active against many Gram-negative rods, chlamydia, mycoplasmas, and rickettsia.

- If effective against one or very few types of bacteria, they are narrow spectrum, eg., Vancomycin is used against certain Gram positive cocci, such as staphylococci and enterococci.

Mechanism of action of antibiotic:

It is characterized by:

- Selective toxicity.
- Antimicrobials must be toxic to the microbe, but not to the host.

Mechanisms of action:

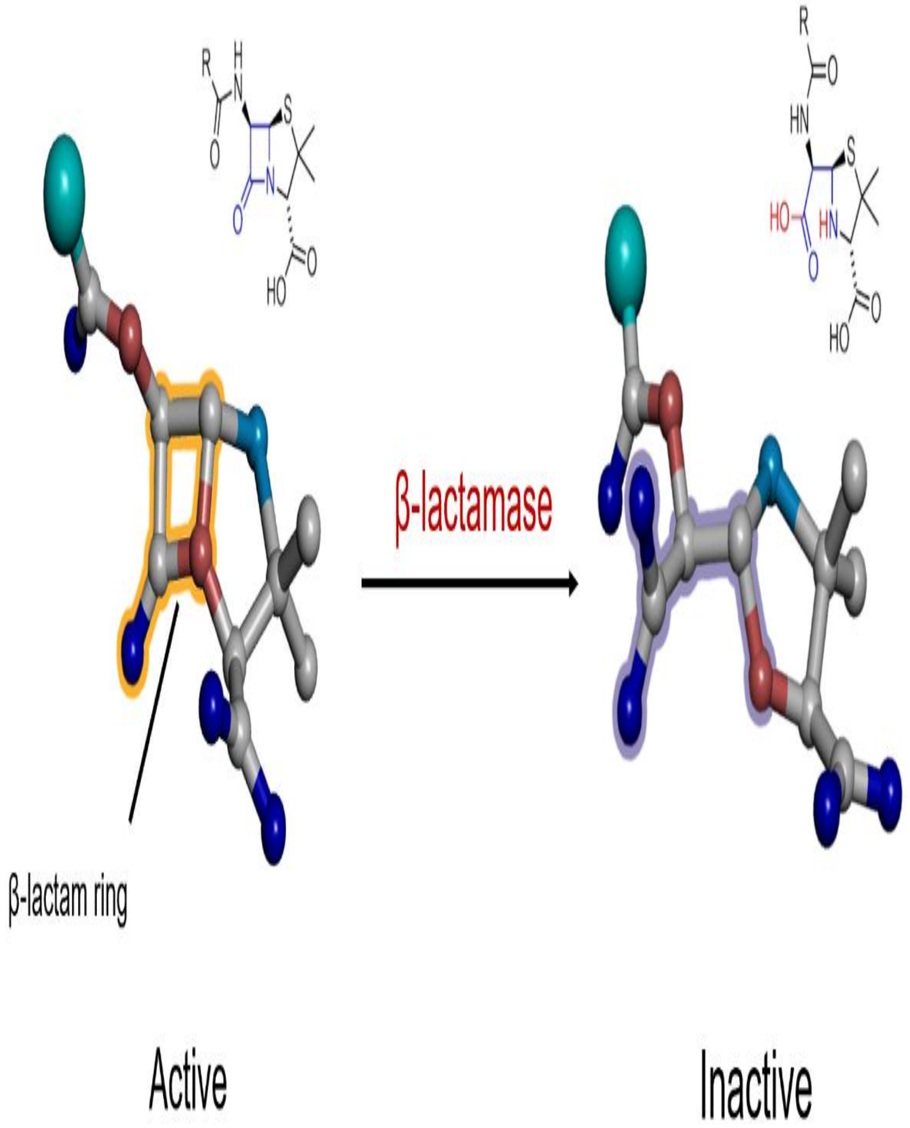
- Cell wall synthesis inhibitors.
- Alteration of cell membrane function.
- Protein synthesis inhibitors.
- Nucleic acid synthesis inhibitors.
- Metabolic pathways inhibitors.

1-Cell Wall Synthesis Inhibitors include:

1-(Penicillin and Cephalosporin) that characteristic by:

- Insoluble in natural form.
- Usually converted to a salt to increase solubility.
- Contains a β-lactam ring that interferes with cell wall synthesis.
- Penicillin is first bound by cellular penicillin binding receptors.
- This binding interferes with transpeptidation reaction.
- This prevents peptidoglycan synthesis.

- Penicillin is bactericidal, but it kills cells only when they are growing.
- In non-growing cells, no new cross-linkages are required and penicillin is inactive.
- Penicillin are therefore more active during the log phase of bacterial cell growth.



2-Vancomycin:

- active against Gram positive cocci, but not Gram negative because too large to pass through outer membrane.
- interferes with peptidoglycan elongation.

3-Cycloserine and Ethionamide :

- Inhibits enzymes that catalyze cell wall synthesis.
- Use for Mycobacterial infections.

2.Alteration of cell membranes (Polymyxin and colistin):

- Polymyxins interact with phospholipids and cause destroys membranes.
- Active against Gram negative bacilli.
- Serious side effects.
- Used mostly for skin & eye infections.

3. Inhibition of protein synthesis (Tetracycline):

- Prokaryotes (70S) and eukaryotes (80S) have a different structure to ribosomes so can use antibiotics for selective toxicity against ribosomes of prokaryotes.

4-Inhibition of DNA/RNA synthesis (Rifampin):

- Binds to RNA polymerase.
- Active against Gram positive cocci.
- Bactericidal for Mycobacterium.
- Used for treatment and prevention of meningococcal.

Side effects of antibiotic:

- Penicillin: Hypersensitivity.
- Cephalosporin: Hypersensitivity, nephritis, hemolytic anemia.
- Tetracycline: Discoloring of teeth.
- Chloramphenicol: Disruption of RBC production.
- Erythromycins: Hepatitis.
- Vancomycin: leukopenia and renal damage.
- Sulfonamides: Hemolytic anemia.

Any Question?

Thank you